

assignment1 - DC circuits - fundamentals

 This is a preview of the published version of the quiz

Started: Mar 22 at 7:01pm

Quiz Instructions

Go over the slides and definitions, print out the equation sheet. Go over the equation sheet.

Make sure to try without looking at the hints (hide them). Remember to convert.

2 attempts. Last one counts .

Work with a group, in person. Collaborative learning is the best



Question 1 6.25 pts

Just as in hydraulic circuits there is water pressure, in electric circuits there is

See slides

☐

voltage.

☐

current.

☐

resistance.



Question 2 6.25 pts

An ampere is a unit of electric

☐

resistance.

☐

voltage.

☐

current.



Question 3 6.25 pts

Power is defined as the energy expended per unit of time. When translated to electrical terms, power is equal to

☐

current multiplied by resistance.

☐

voltage divided by time.

☐

none of the above

☐

current multiplied by voltage.

☐

current divided by time.



Question 4 6.25 pts

What current is flowing in a wire if 0.67 C of charge pass a point in the wire in 0.30 s ?

hint: see slides for definition of current and equation

☐

2.2 A

☐

0.30 A

☐

0.20 A

☐

0.67 A



Question 5 6.25 pts

A charge of 12 C passes through an electroplating apparatus in 2.0 min . What is the average current in the apparatus?

hint: convert min to seconds.

☐

0.10 A

☐

6.0 A

☐

1.0 A

☐

0.60 A



Question 6 6.25 pts

In units of measurement, power in watts is equal to

See my slides if you are not sure.

☐

none of the above

☐

amperes x ohms.

☐

amperes/second.

☐

volts/second.

☐

amperes x volts.



Question 7 6.25 pts

Which statement is correct?

☐

resistance flows in a circuit

☐

all of the above

☐

charge flows in a circuit

☐

voltage flows in a circuit



Question 8 6.25 pts

The unit of electrical resistance is the

☐

none of the above

☐

ohm.

☐

ampere.

☐

volt.



Question 9 6.25 pts

One joule per coulomb is a unit of

☐

power.

☐

energy.

☐

resistance.



voltage.



current.



Question 10 6.25 pts

A coulomb of charge flowing in a bulb filament powered by a 6-volt battery is provided with



6 ohms.



6 watts.



6 newtons.



6 joules.



6 amperes.



Question 11 6.25 pts

How much charge must pass by a point in a wire in 10 s for the current in the wire to be 0.50 A?



0.050 C



5.0 C



2.0 C



20 C



Question 12 6 pts

TO make a potato battery, in addition to the potato, load and connecting wire, you could use



1 copper wire and 1 brass nail



2 plastic rulers



2 brass nails



2 copper wires



Question 13 4 pts

A positive ion is formed when (for sure)

☐

a negative ion loses an electron

☐

a neutral atom gains an electron

☐

It depends on the element

☐

a neutral atom loses an electron



Question 14 4 pts

A battery supplies _____ - and wall outlet supplies _____

☐

DC current, AC current

☐

DC current, DC current

☐

It depends on the voltage

☐

AC current, DC current



Question 15 6 pts

In class, I showed you that the voltage of the wall outlet was about:

☐

9V

☐

120V

☐

1.5V

☐

12V



Question 16 6 pts

2 AA batteries are connected in series. The voltage is now

hint: see the slides for AA batteries

☐

9V

☐

6V



1.5V



3V



Question 17 6 pts

In a DC circuit (as shown in class) with wires, power supply and loads
charge carriers are the particles that "flow" in a circuit.

(not according to Franklin. But in reality)



electrons are the charge carriers



frogs are the charge carriers



protons are the charge carrier



ions are the charge carriers

Not saved

Submit Quiz